Name: ____________________ TOTAL = /31

THE NERVOUS SYSTEM

Topic 1 Review: Neurons & Electrochemical Impulses

1) Match each term below to its appropriate definition... **(11)**

A. Glial cells	_____ cannot be re-generated after injury
B. Dendrites	_____ where neurotransmitters are stored
C. Nodes of Ranvier	_____ a state of balance within the body
D. Axon terminal	_____ composed of myelinated neurons
E. Sensory neurons	_____ a neuron at rest
F. Motor neurons	_____ gaps in the myelin sheath
G. Homeostasis	_____ support neurons structurally & nutritionally
H. White matter	_____ a neuron that has undergone an action potential
I. Grey matter	_____ receive signals from adjoining neurons
J. Polarized	_____ relay signals from the brain to the rest of the body
K. Depolarized	_____ afferent neurons

2) A motor neuron is compared to an interneuron in the brain. Which neuron is capable of conducting electrical impulses the most rapidly? Explain why. **(2)**

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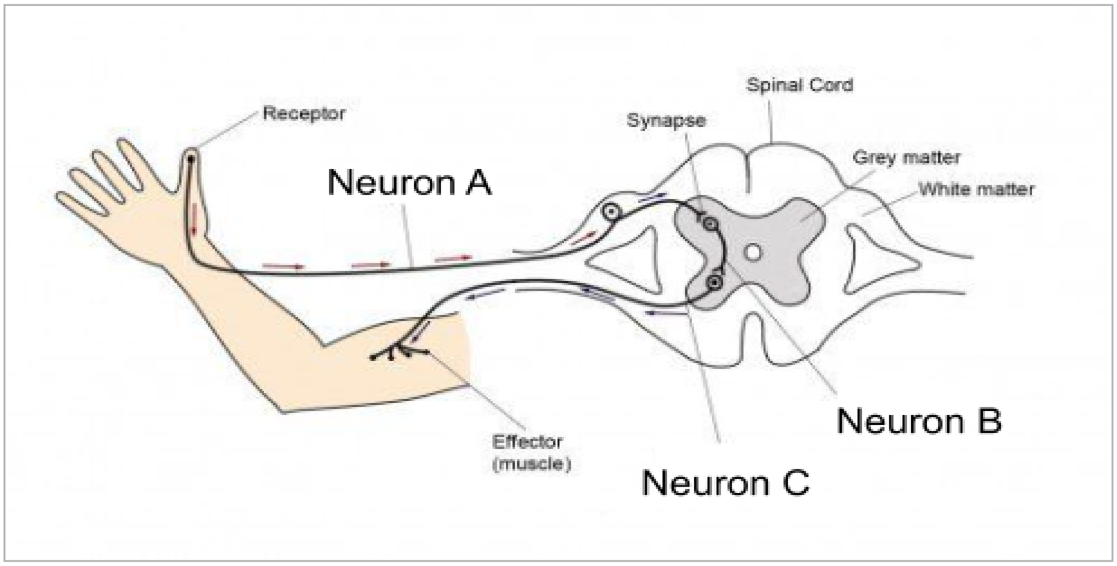
3) Consider the following diagram of a reflex arc:



Discuss some symptoms an individual would experience if the group of neurons labelled "Neuron C" in the diagram above were impaired. **(2)**

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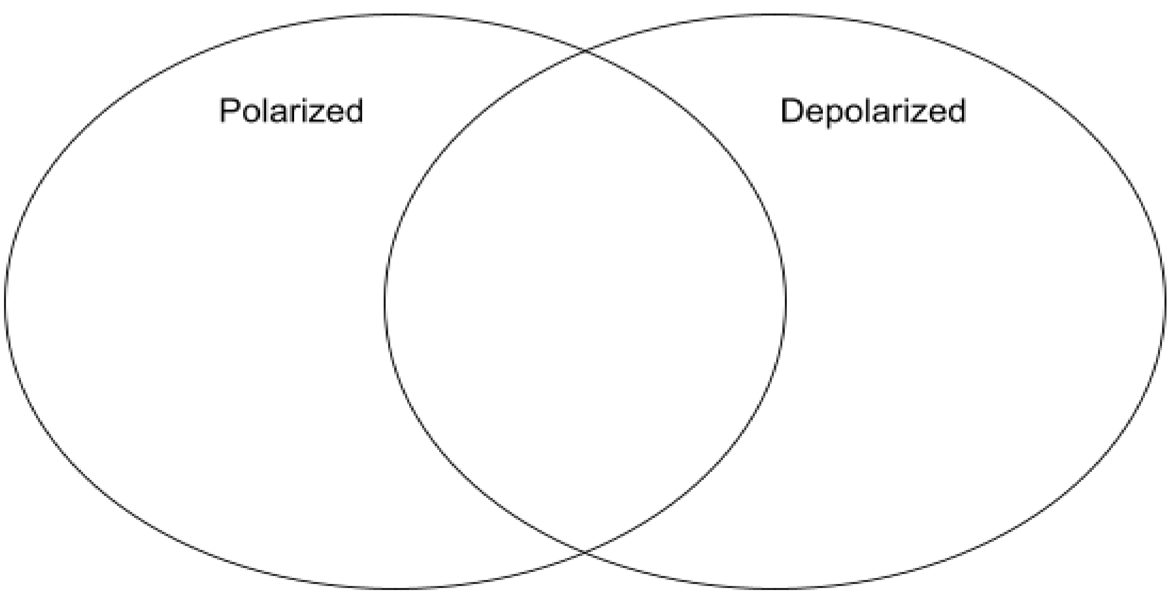
4) Complete the following Venn Diagram comparing polarized vs. depolarized neurons. Include two similarities and two differences for full marks. **(4)**



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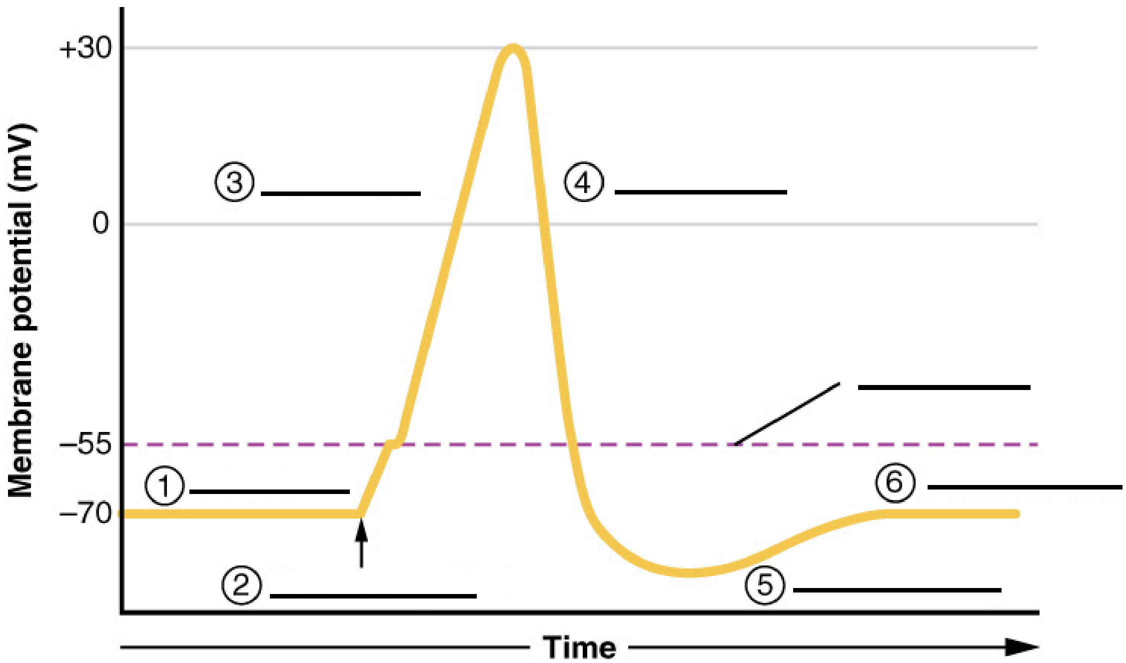
5) Consider the following graph of the membrane potential of a neuron:



a. Label each component of the graph above. **(3)**

b. At what point on the graph above are Na^+ gates open? **(1)**

c. At what point on the graph above are K^+ gates open? **(1)**

d. At what point on the graph above is the neuron said to be "hyperpolarized"? **(1)**

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6) Consider the "all-or-none" response of an electrical impulse. Why do we experience pain as a result of certain stimuli but not others? **(2)**

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7) Explain the difference between excitatory and inhibitory neurotransmitters. Provide an example of each. **(2)**

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8) Insecticides work by blocking the release of cholinesterase, causing muscles to remain in a state of contraction. Explain the role of cholinesterase in synaptic transmission, and hypothesize how an overproduction of cholinesterase would affect muscle functioning. **(2)**

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